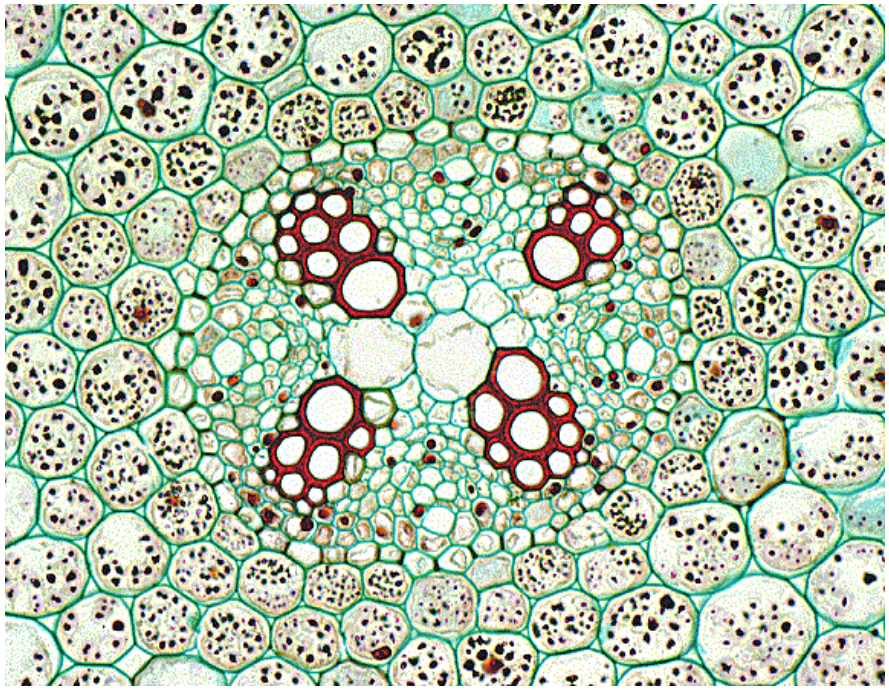
Plant Tissues

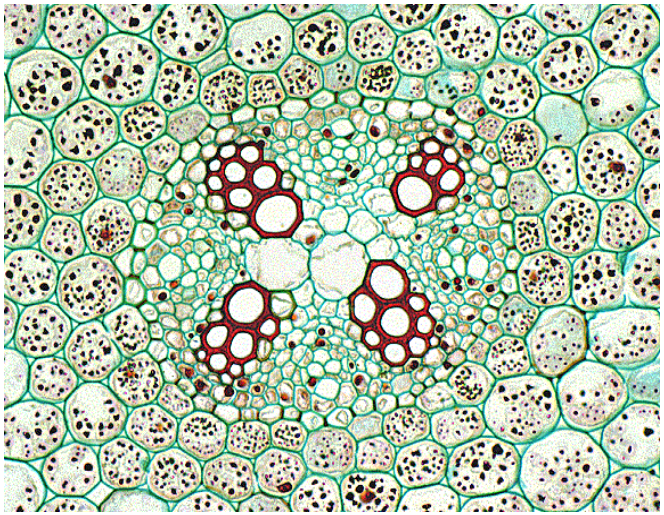
Examples of the Different Types of Plant Cells

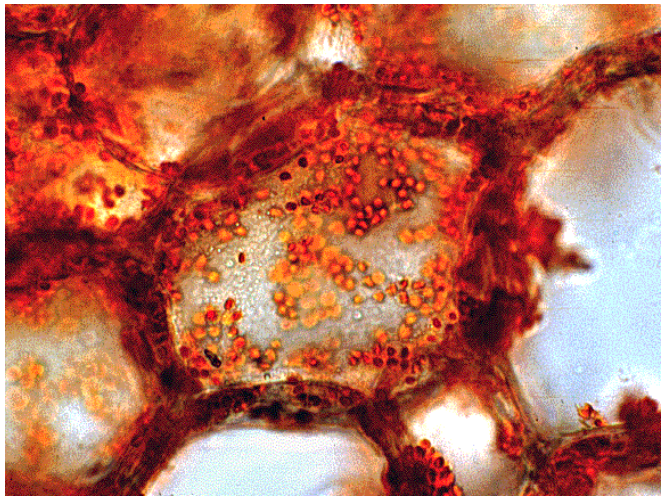
- Parenchyma
- Collenchyma
- Sclerenchyma
  - fibers and sclerids
- Vessels
- Tracheids
- Sieve tube members
- Companion cells
- Egg (ovum)
- Sperm



The Parenchyma Cell

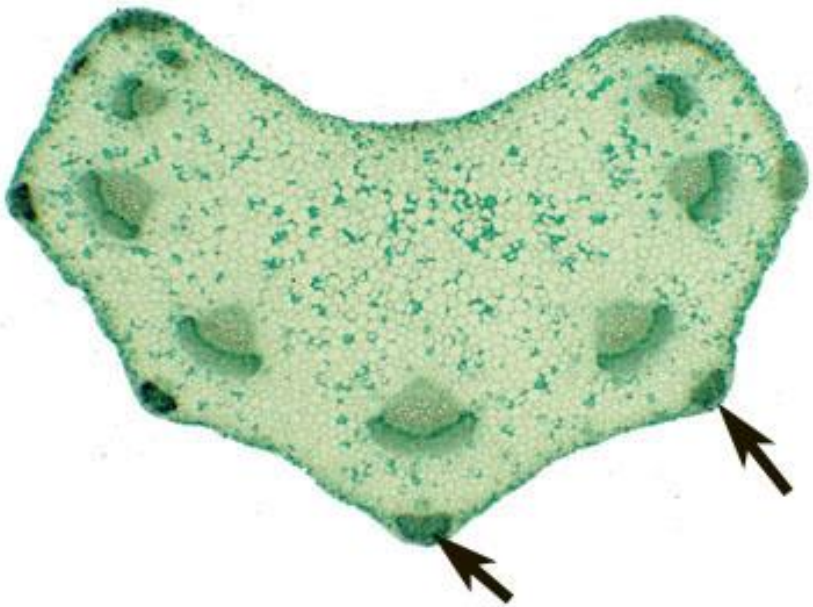
- The generic plant cell
- Most abundant and versatile
- Function:
  - storage
    - flesh of fruit
    - storage in roots and seeds
  - basic metabolism
- They can change and become other cells (differentiation)
- **Alive at maturity**
- Primary cell wall

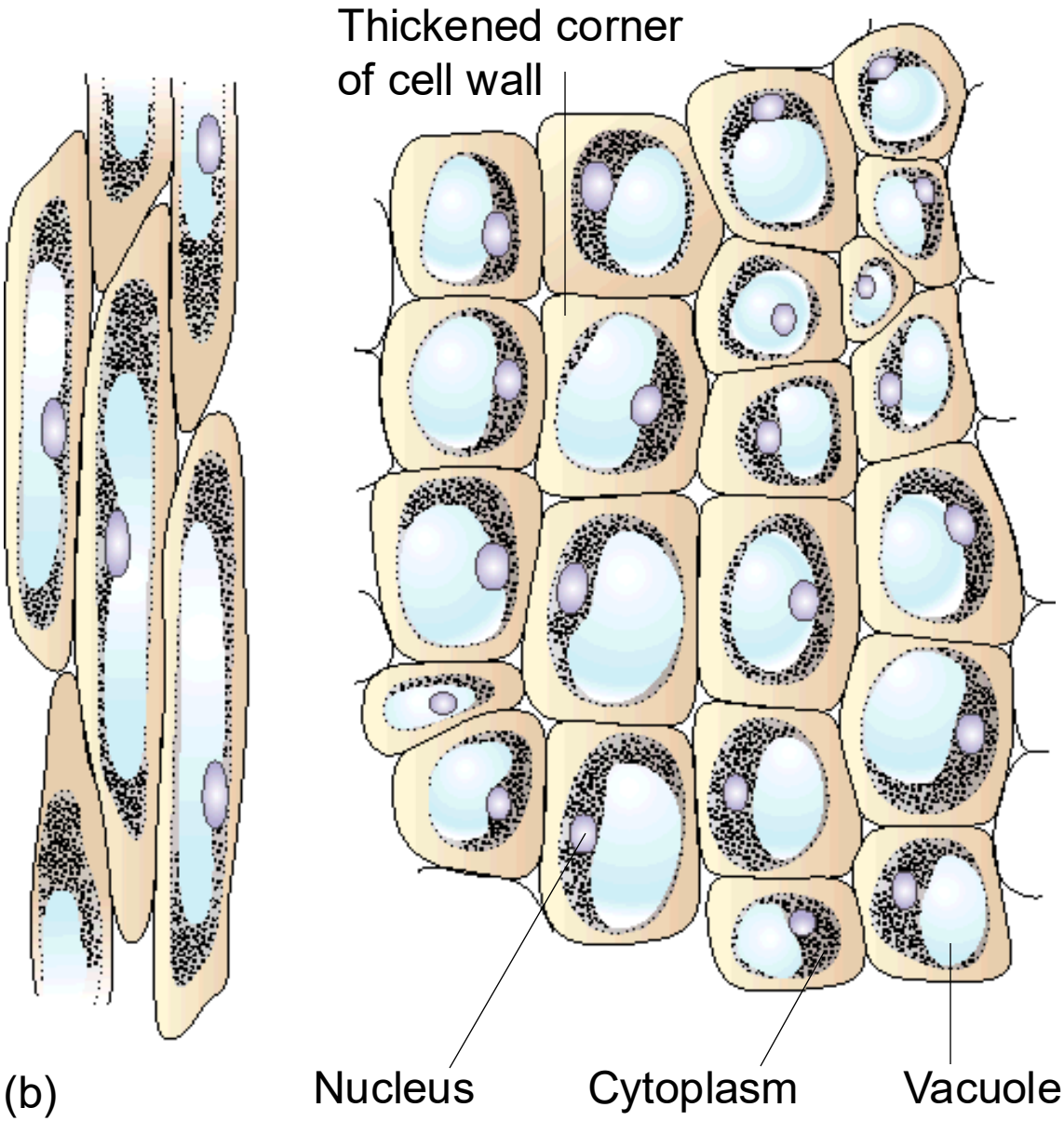




The Collenchyma Cell

- “glue” cells
- Flexible support
- Found in growing shoots, stems, & leaves
- Differentiate from parenchyma cells
- **Alive at maturity**
- Primary and secondary cell wall
- Ex: resilient strings found in celery





© 2002 Brooks/Cole - Thomson Learning

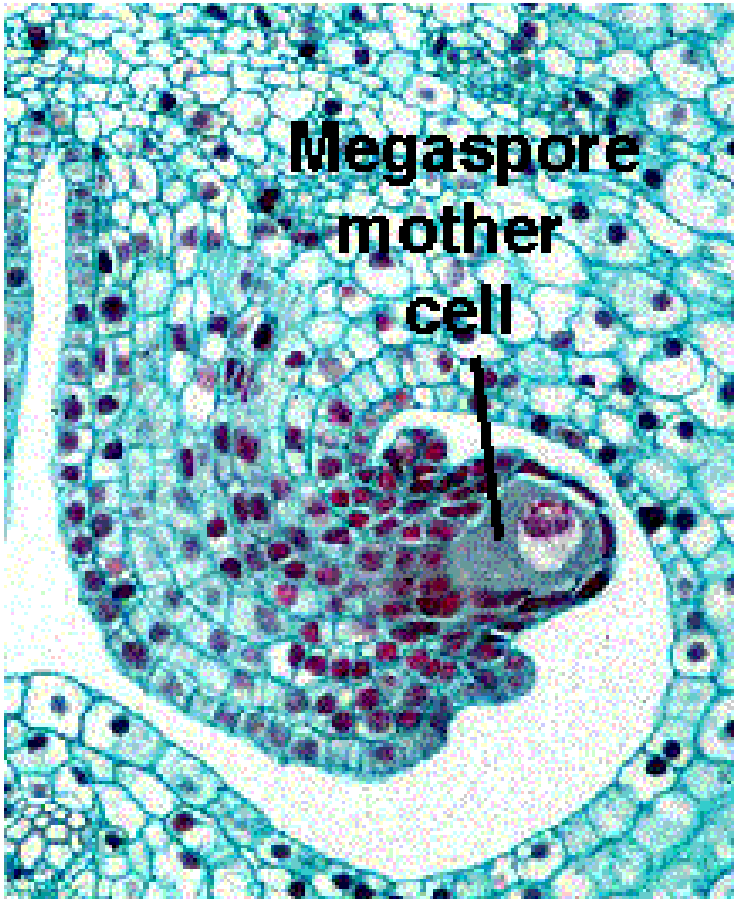
Sclerenchyma

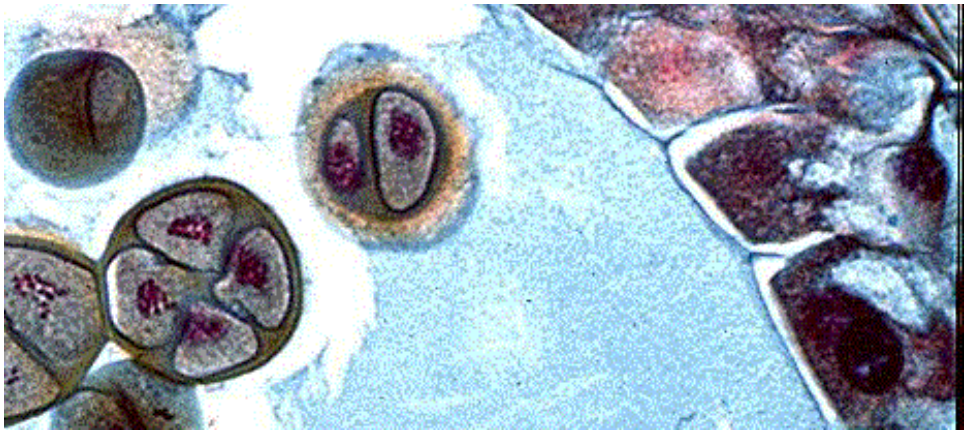
- Rigid cells
- Function: support and strengthen non extending regions of a plant like mature stems
- Thick, non stretchable secondary cell walls
- **Dead at maturity**
- Two Types:

1. Fibers
   - long, slender, occur in strands
   - Cotton
   - Hemp
   - Flax (Linen)

2. Sclerids
   - short, varying shape, occur in groups
   - Gristle in pears

Plants Eggs and Sperm





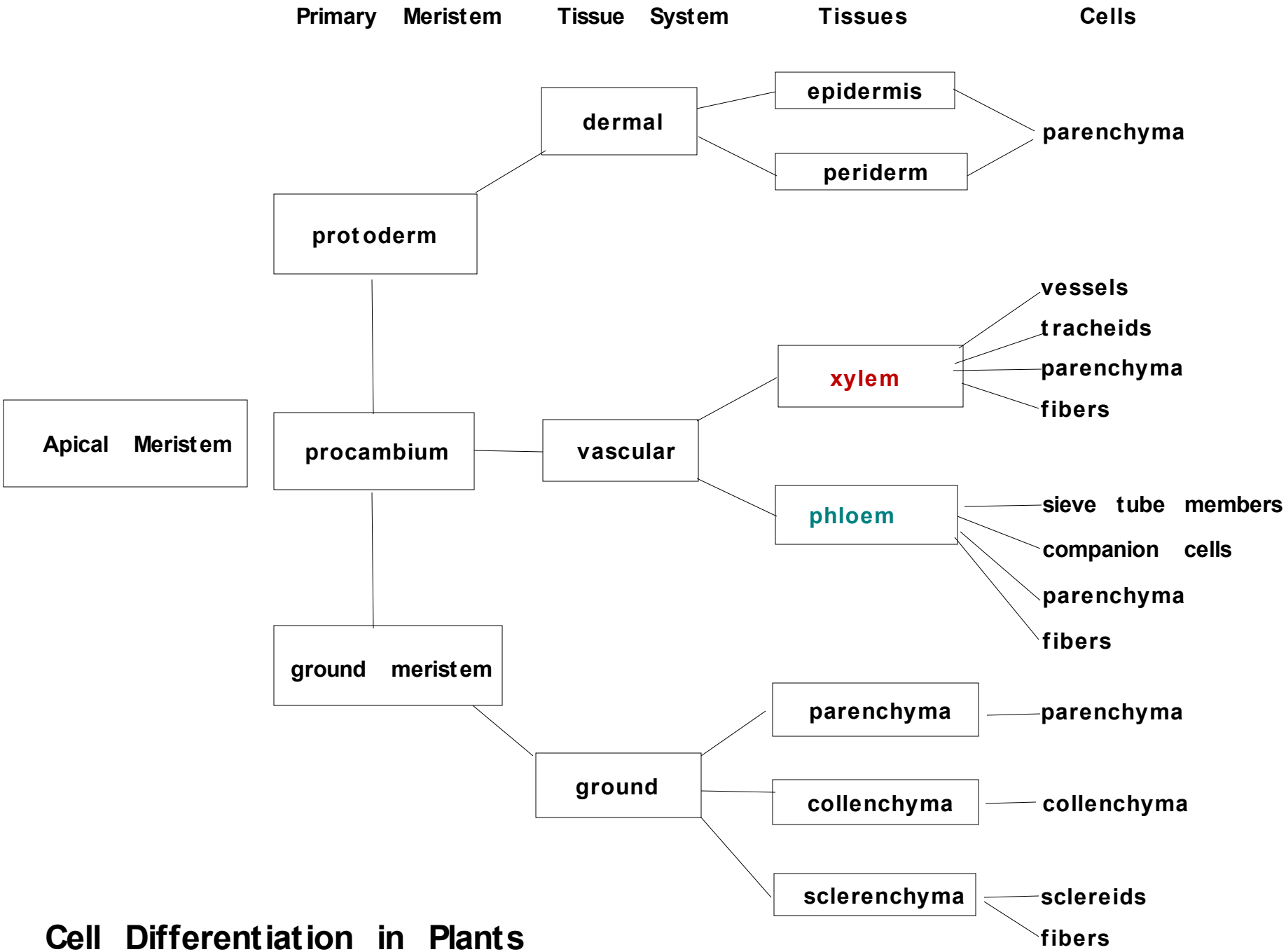
Cells→Tissues

- Like animal cells, plant cells are organized into tissues

- Tissues are groups of cells that form a structural and functional unit
  - Simple tissues have one cell type
  - Complex tissues have two or more cell types

Tissues→ Tissue System

- Tissues are organized into tissue systems
- Vascular plants have three tissue systems
  - Ground tissue
    - Energy transformation, storage and support
  - Vascular tissue
    - Water and nutrient conduction system
  - Dermal tissue
    - Covering for the plant body



Cell Differentiation in Plants

- **Building a House**

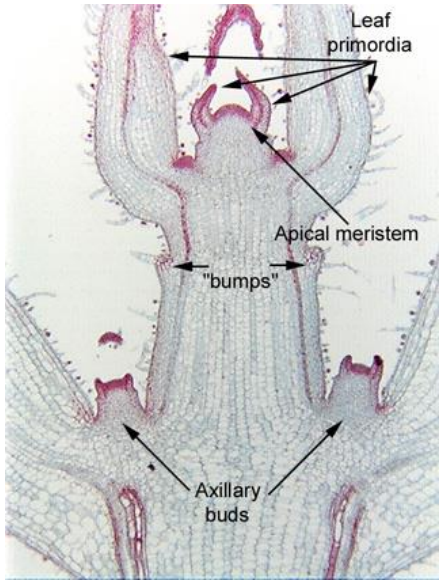
1. Foundation is laid
2. Construction of the frame
3. Installation of plumbing, heating, etc
4. Waterproof walls and roof
5. Food stored in appropriate places

- **Building a plant**

1. Meristems give rise to all tissues
2. Three tissue systems give rise to the major organs of a plant
3. Installation of the vascular tissue (plumbing)
4. Installation of dermal tissues
   - covering, skin
5. Installation of ground tissue
   - parenchyma

Meristems

- Meristems are regions of cell division
- Main difference between animals and plants
- Birds and mammals stop growing at maturity, although certain cells get replenished (skin)
- Plants have the ability to keep growing their entire lives
- Meristems make it possible to grow a new plant from a cutting
- Growth in plants is the counterpart (to some extent) to movement in animals

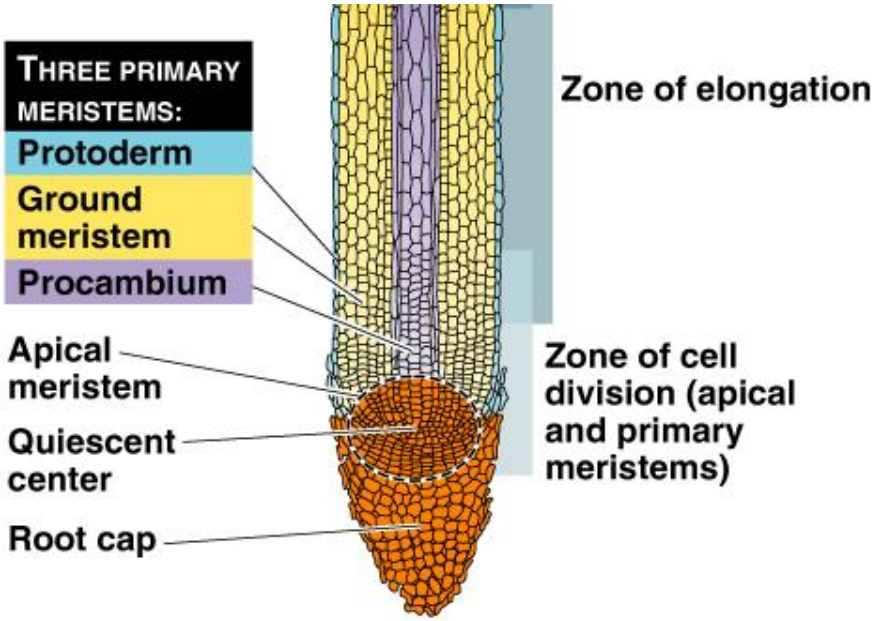


Apical Meristems

- Shoot apical meristem
- Root apical meristem
- Found at the tips of roots and shoots
- Increase the length of a plant

Apical meristems give rise to:

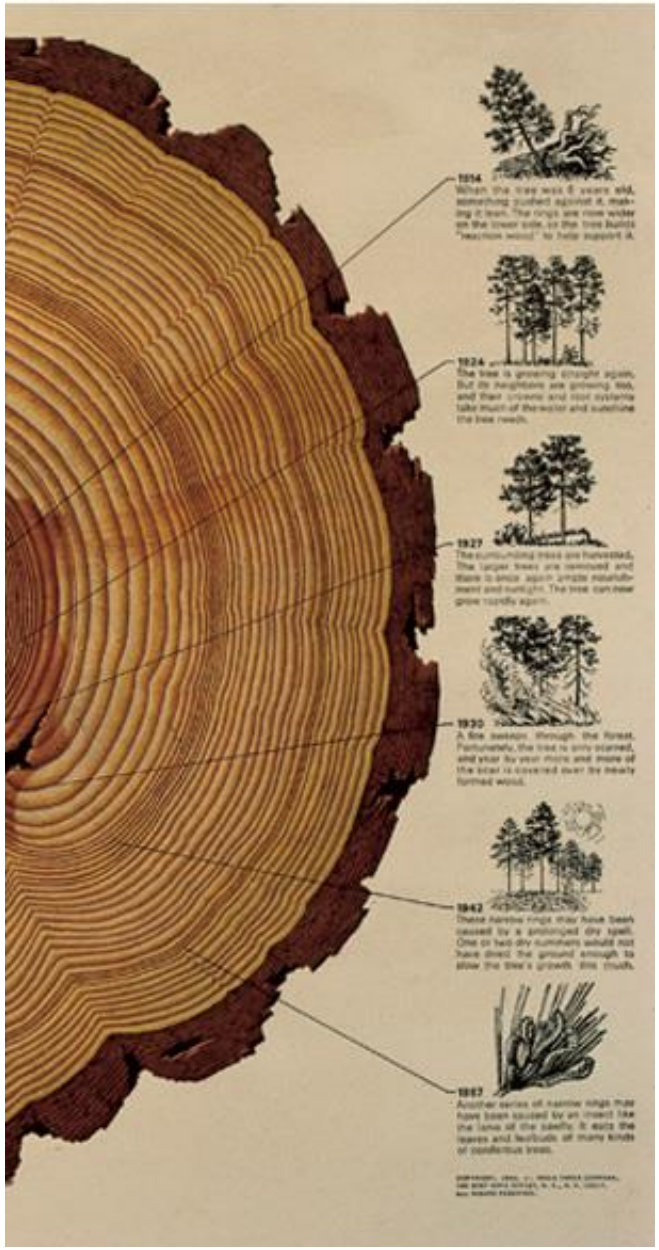
- Three primary meristems
  1. Ground meristem
     - Give rise to ground tissues
  2. Protoderm
     - Give rise to “skin” coverings
  3. Procambium
     - Give rise to plumbing of the plant



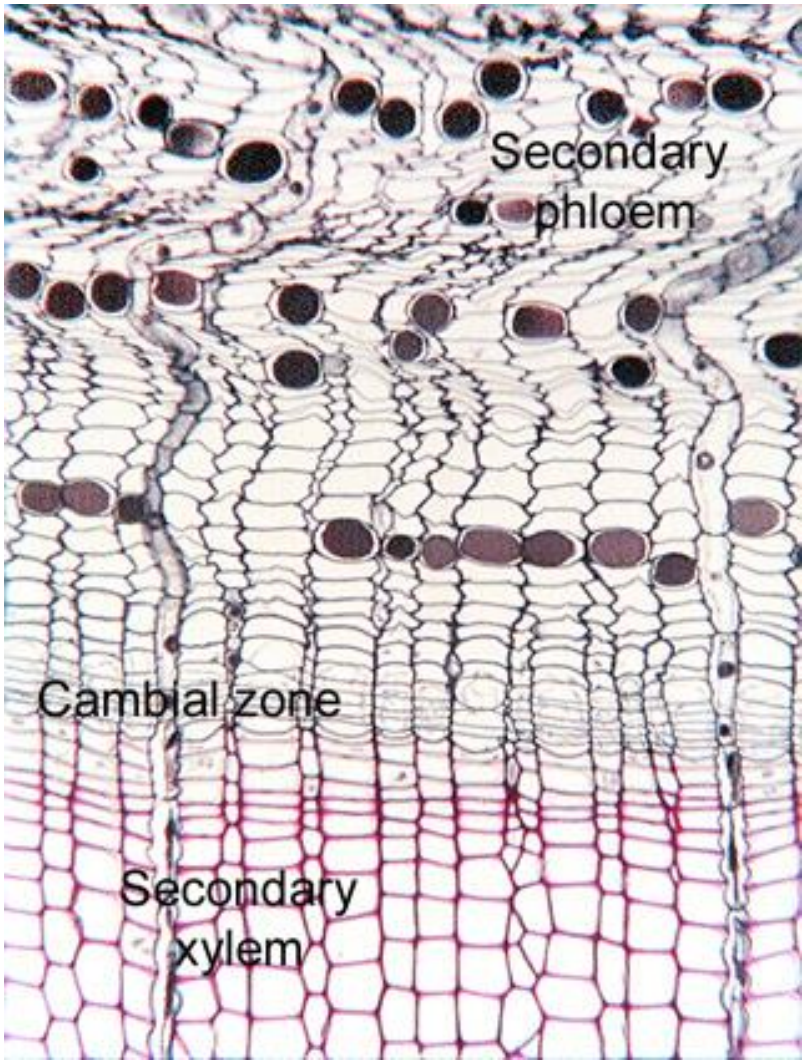
Other Meristems:
Lateral meristems

- **Secondary Growth**
- **Add girth (width) by producing wood and bark**

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.

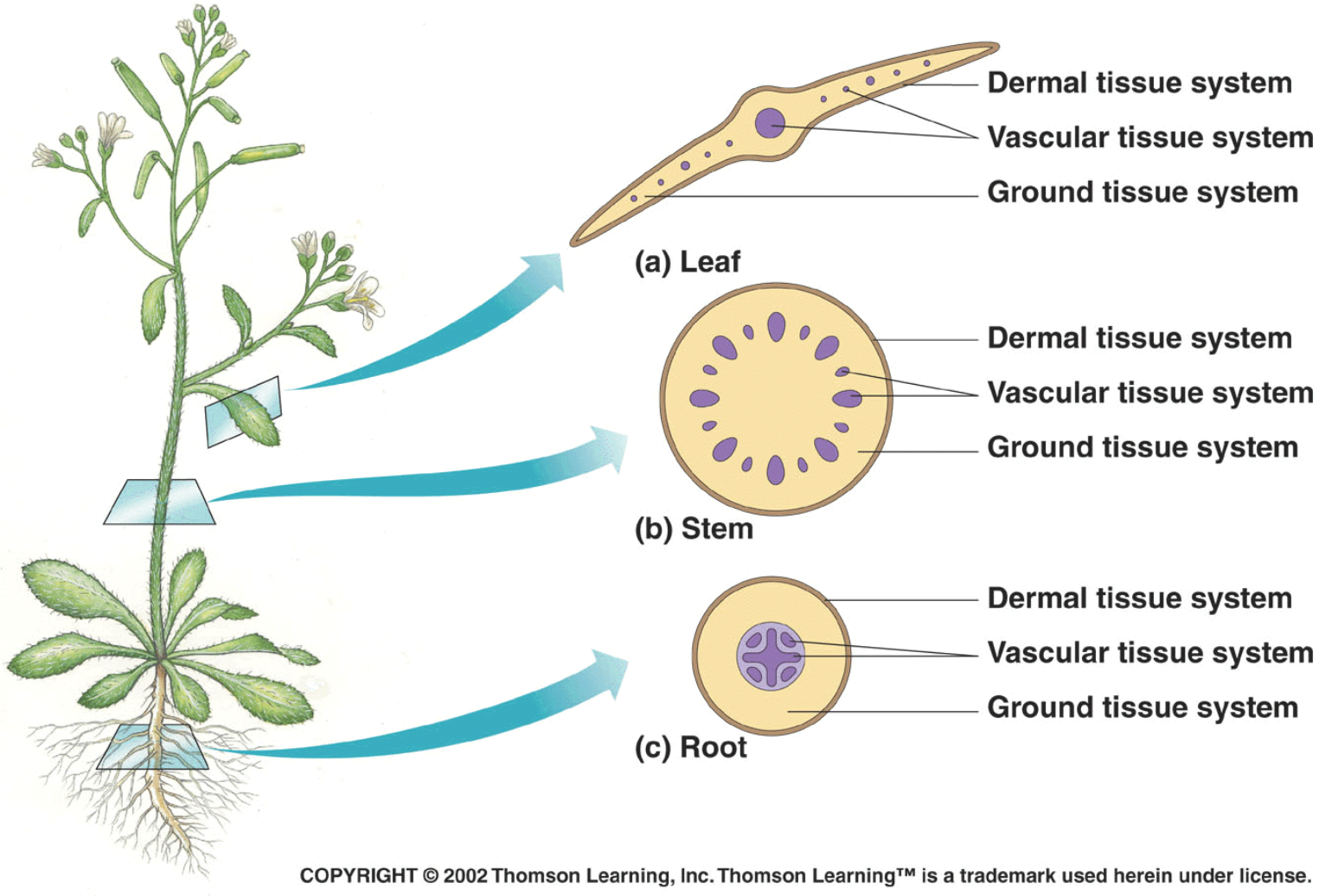


Other Meristems: Intercalary Meristems



- Why grass keeps growing back after you cut it.
- Grasses and related plants do not have lateral meristems
- Intercalary meristems add to the plants length

The Tissue Systems Give Rise to the Organs of a Plant

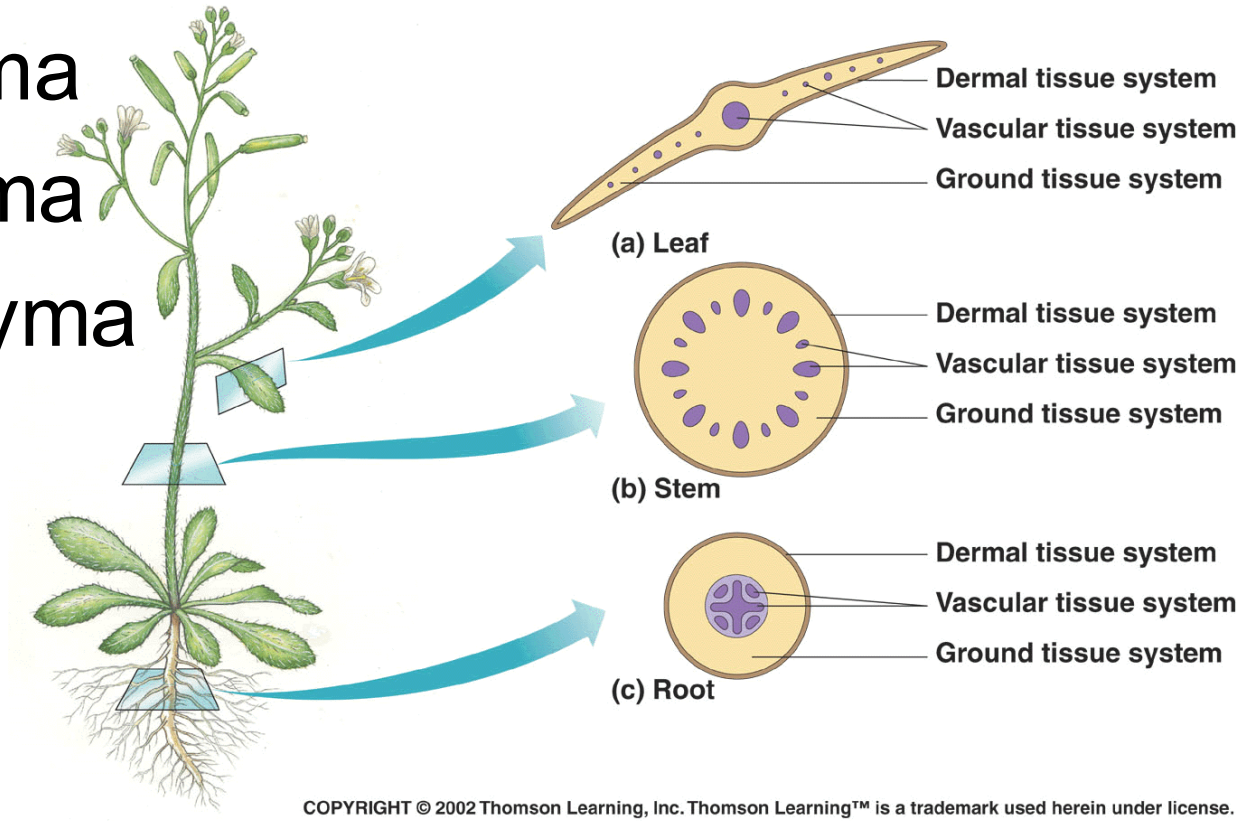


COPYRIGHT © 2002 Thomson Learning, Inc. Thomson Learning™ is a trademark used herein under license.

Ground Tissue

- Tissues that "ground" the plant
- Fundamental tissues of a plant

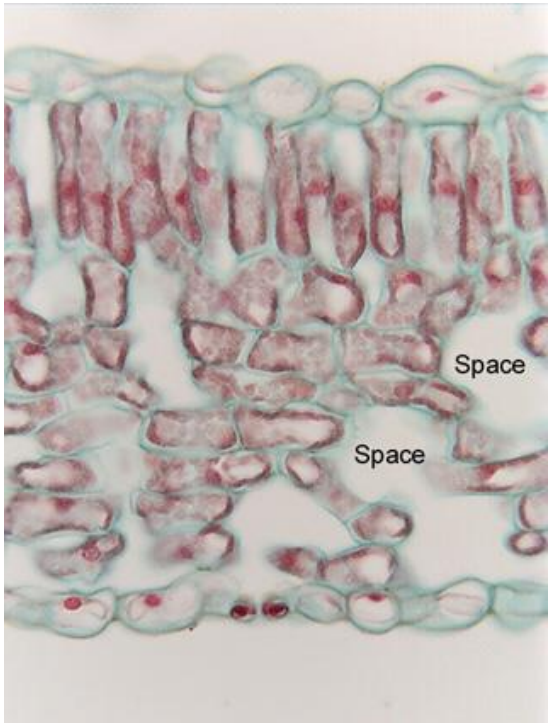
1. parenchyma
2. collenchyma
3. sclerenchyma



COPYRIGHT © 2002 Thomson Learning, Inc. Thomson Learning™ is a trademark used herein under license.

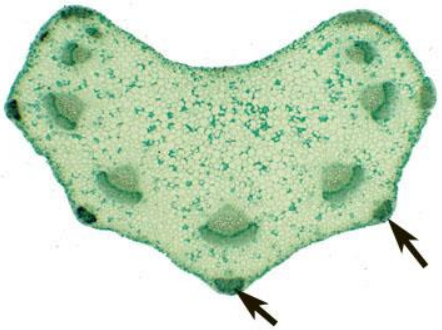
Ground Tissues: Parenchyma

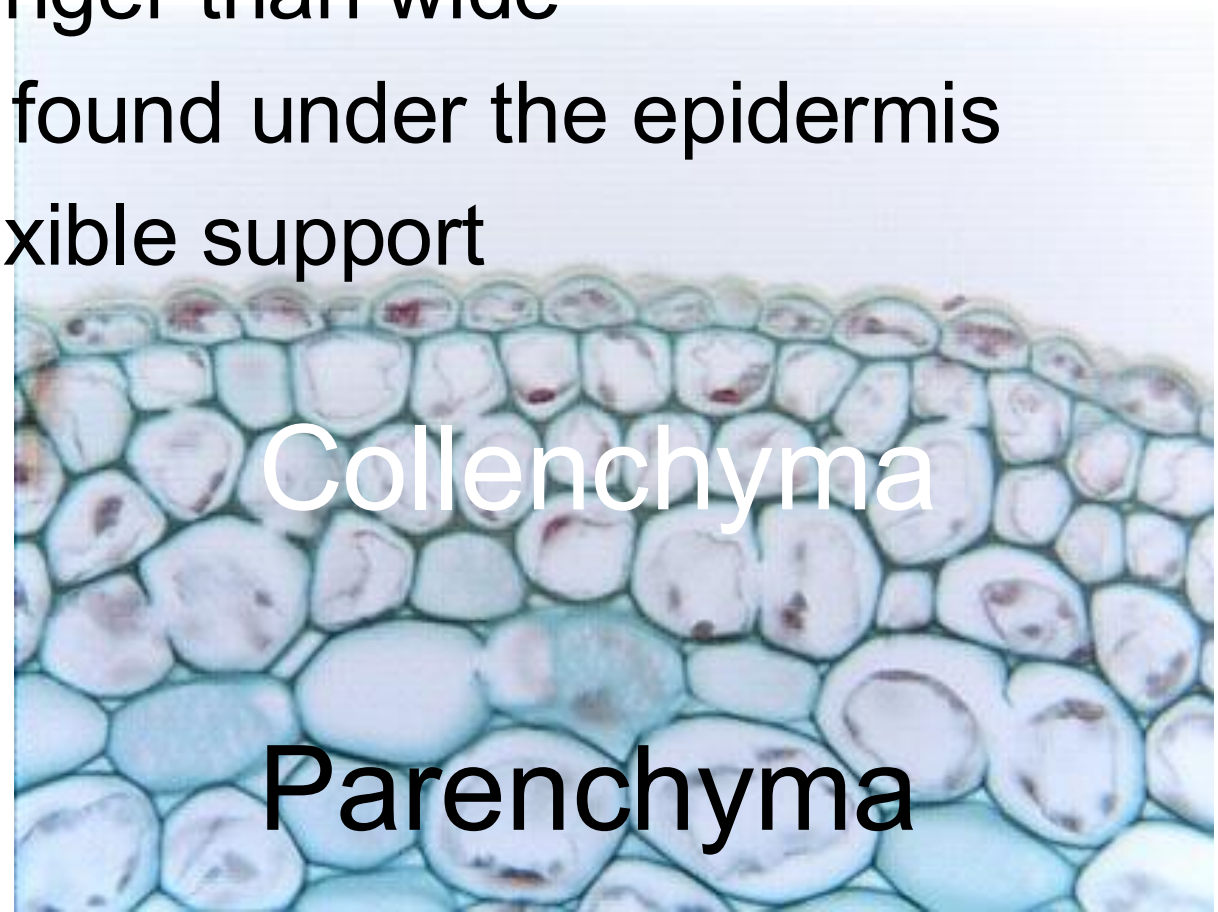
- cells assume various shapes and sizes
- contain starch grains or other storage granules
- oils
- tannins (tanning or dyeing substances)
- crystals
- various other secretions



Ground Tissues: Collenchyma

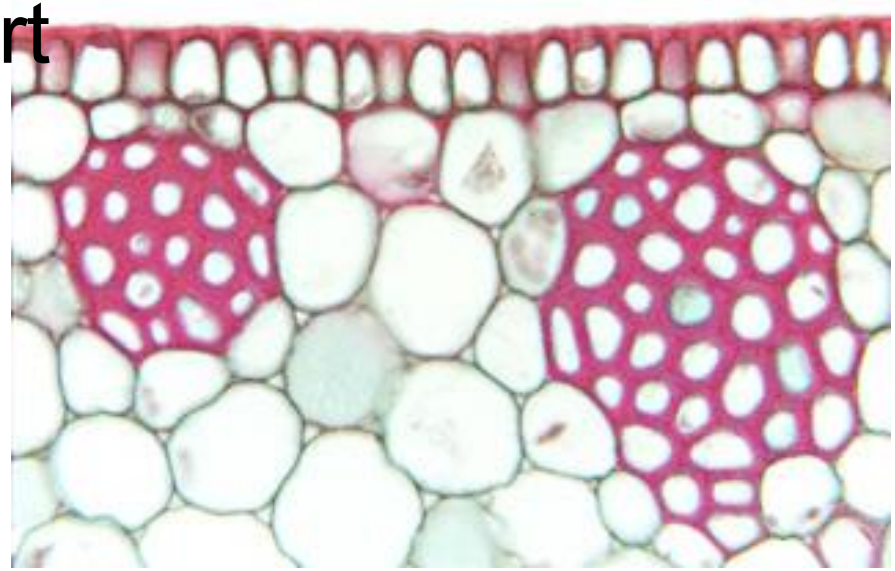
- are alive at maturity
- cell walls are thicker and more uneven
- typically longer than wide
- tend to be found under the epidermis
- provide flexible support





Sclerenchyma

- Tissues (cells) have thick, tough secondary cell walls made of lignin
- Dead at maturity
- Function in support
  1. sclerids
  2. fibers



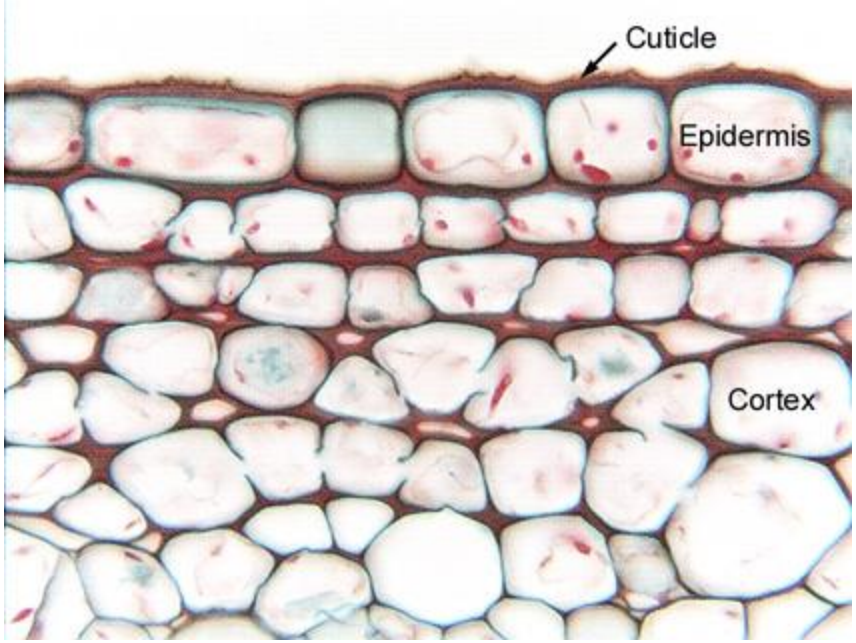
Fibers are used for clothing

- Cotton
- Linen
- Bamboo
- Hemp

The Dermal and Vascular Tissue Systems are made of Complex Tissues

- Tissues that are made of more than one cell type
- Dermal Tissues
  - epidermis
  - endodermis
- Vascular tissues
  - Xylem
  - Phloem

Dermal Tissues

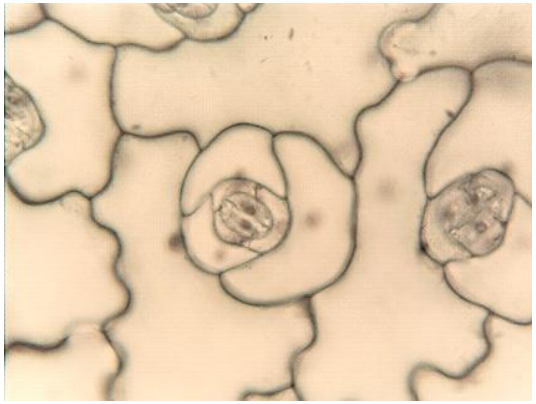


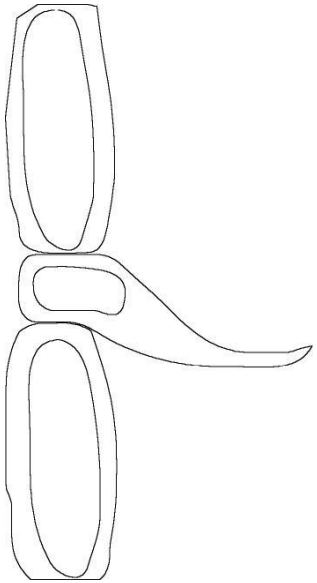
- dermal tissues cover the organs of a plant. They are analogous to skin.
- epidermis
- outermost layer of cells
- usually one cell layer thick
- epidermal cells have a layer of cutin (cuticle)
  - resistant to bacteria and other disease organisms

Cuticle Wax

- Wax Myrtle
- *Myrica cerifera*
- waxes obtained from boiling the leaves is used to make bayberry candles

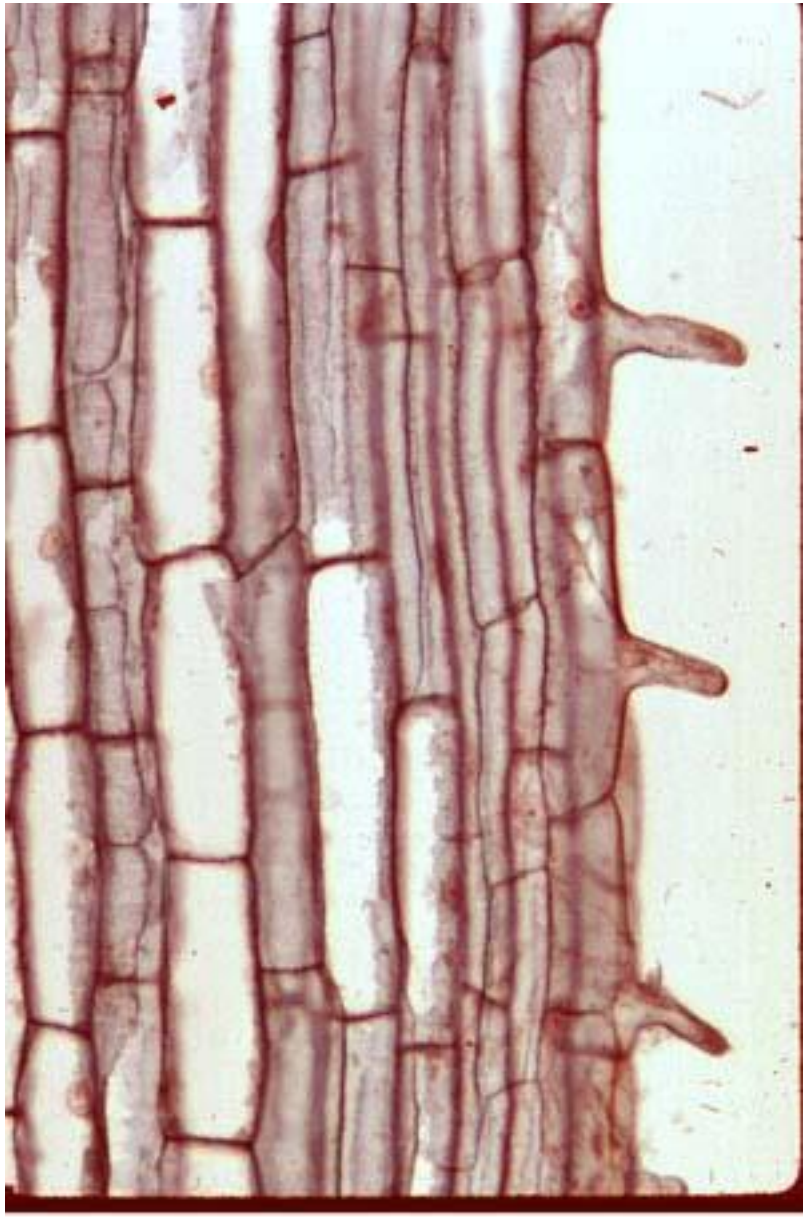
Epidermis is made of many types of cells



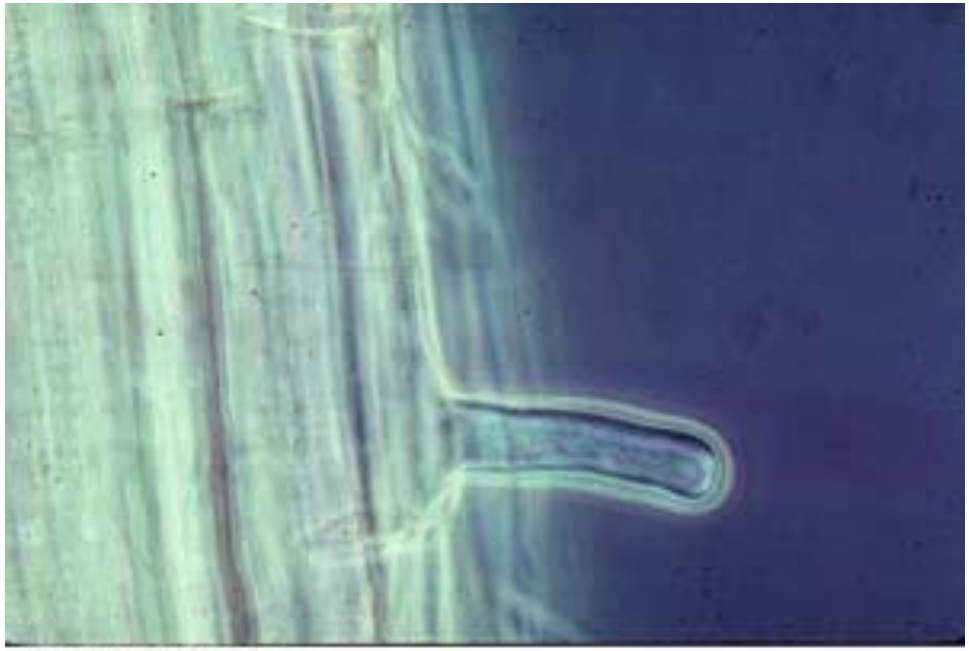


1. **Guard cells form stomata**
2. **Trichomes**
   - Outgrowths – ‘hairs’
   - Some leaf trichomes use to keep animals away
   - Some used to eliminate salt
   - Some thought to increase reflection to reduce stem/leaf temperature
   - *Root hairs* are a type of trichome, increase surface area to increase uptake of nutrients
3. **Glands**

Root Hairs are extensions of epidermal cells



Root hairs.



Live root hair.

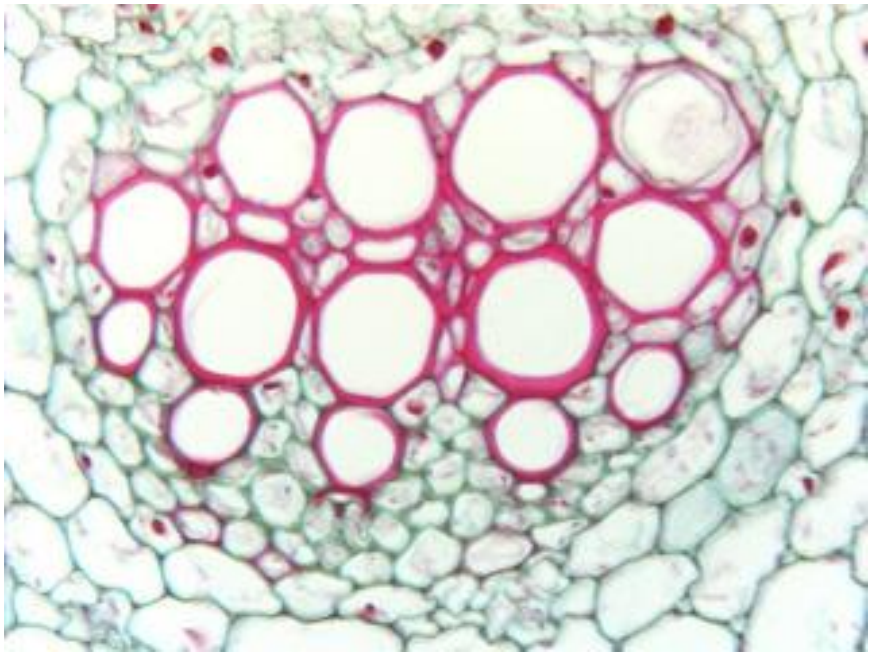
Dermal Tissues: Epidermis

- Glands secrete substances that protect the plant
- secrete nectar
- digestive glands
- Sundews
- trigger hairs of a Venus Flytrap

Complex Tissue: Vascular Tissues

- Xylem Tissue
  - transports water and dissolved substances through the plant to all its organs

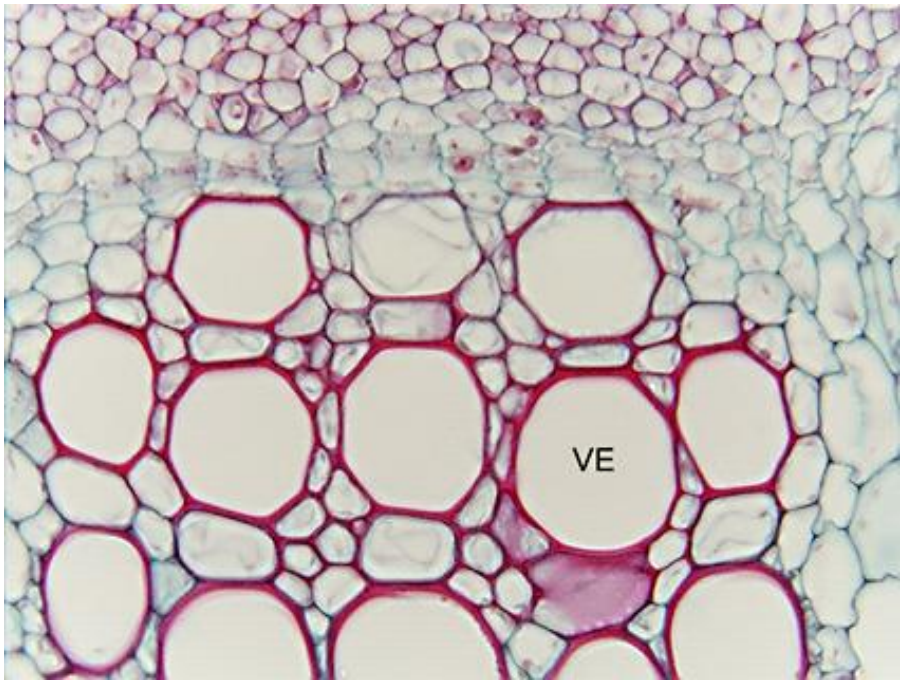
1. parenchyma cells
2. fibers
3. tracheids
4. vessels
5. rays



Xylem

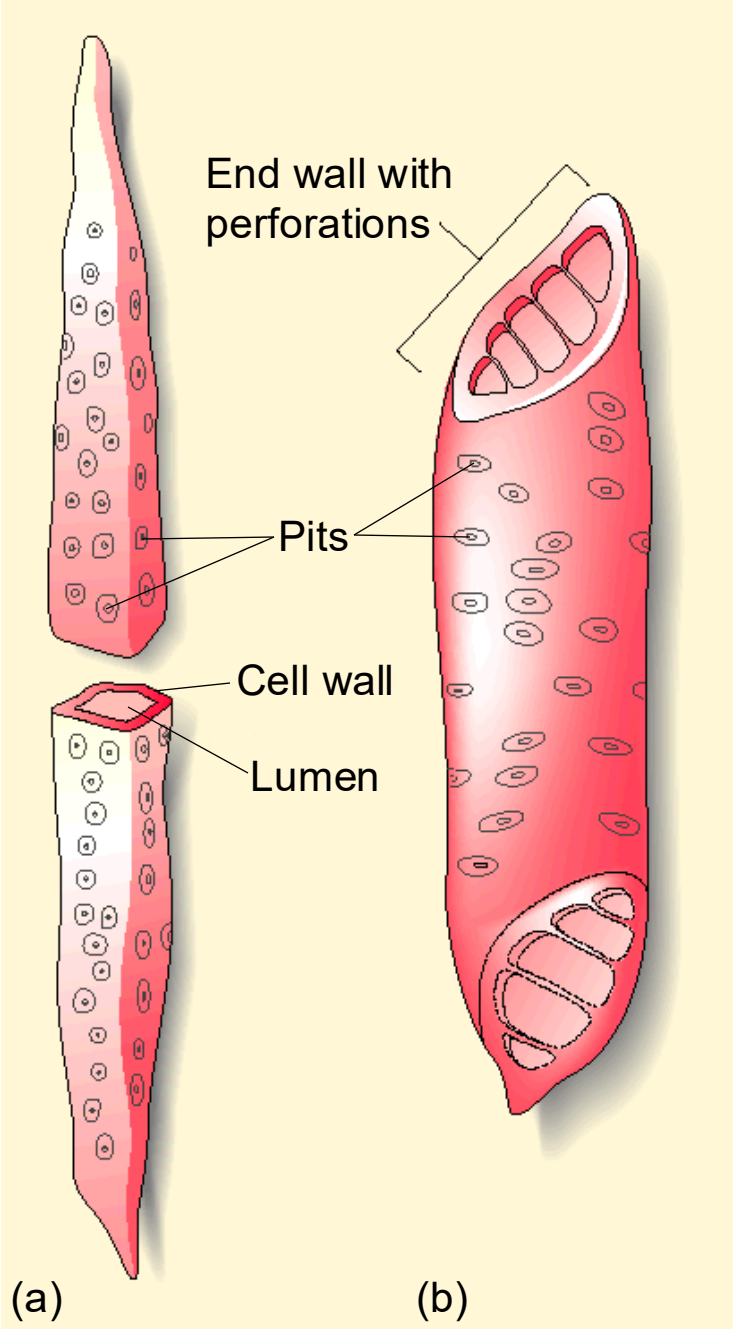
- Tracheids
- Have thick secondary cell walls
- dead at maturity
- tapered at the end
- contain pits
  - areas where no 2° cell wall is laid down

Xylem



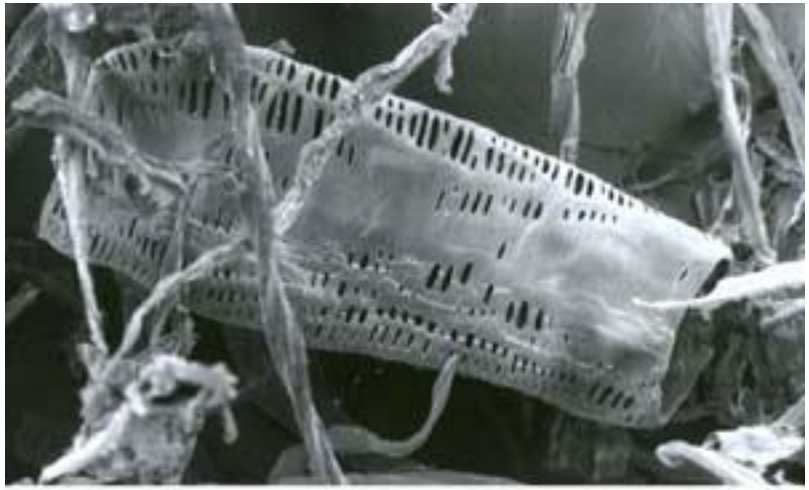
- Vessels
- Have thick secondary cell walls
- dead at maturity
- long tubular cells with grate-like openings at the end
- contain pits
  - areas where no 2° cell wall is laid down

Tracheid

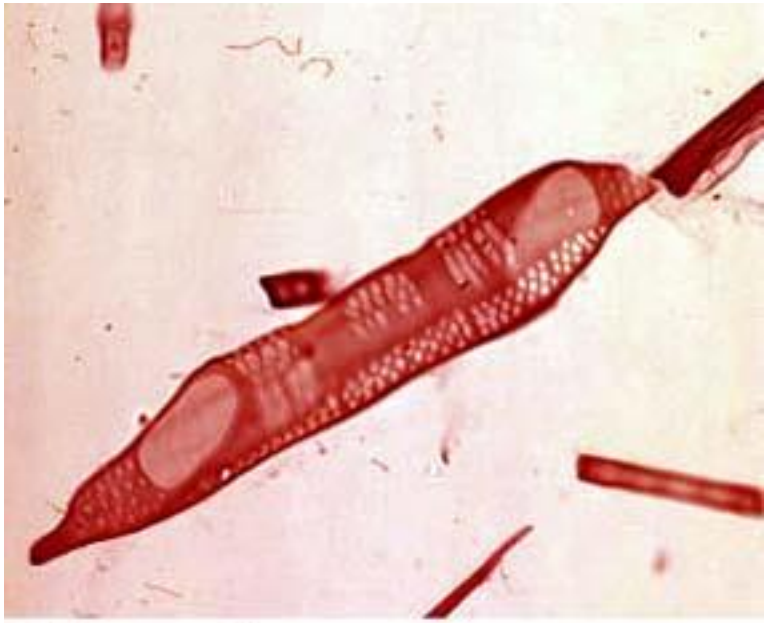


Vessel

© 2002 Brooks/Cole - Thomson Learning

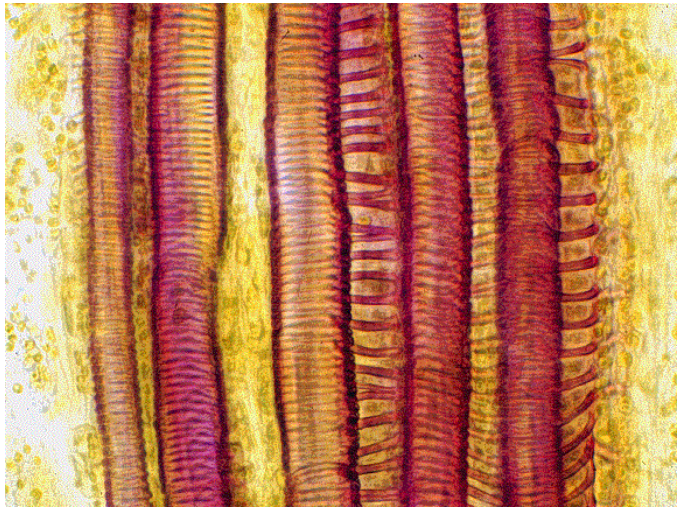


Cucurbita stem maceration- xylem vessel with simple perforation plate

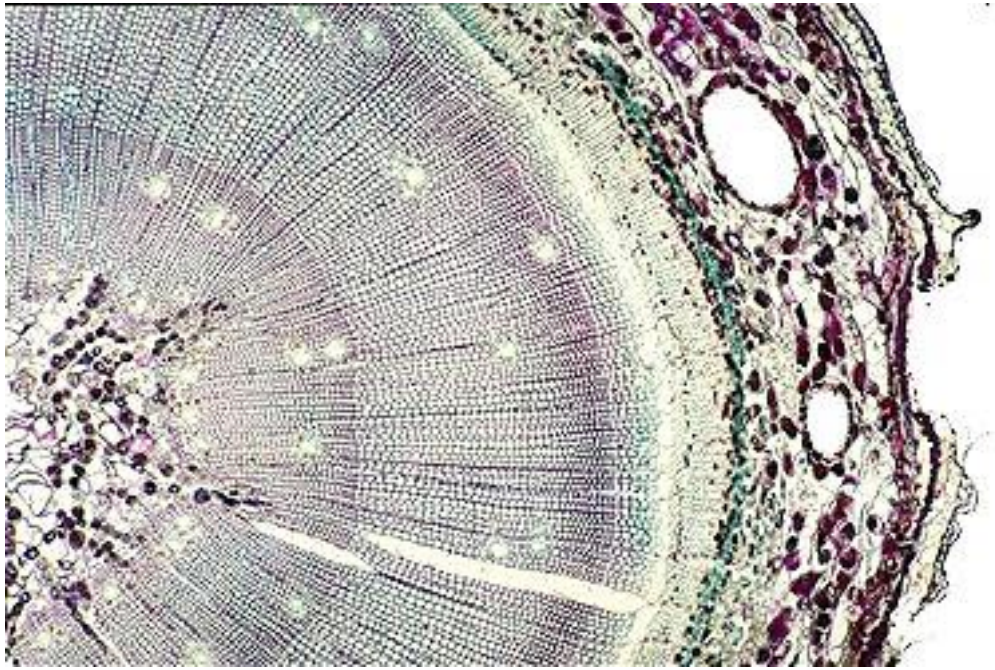


Salix vessel.

Quercus macerated xylem with vessel elements.



Many cone bearing trees and other non-flowering plants

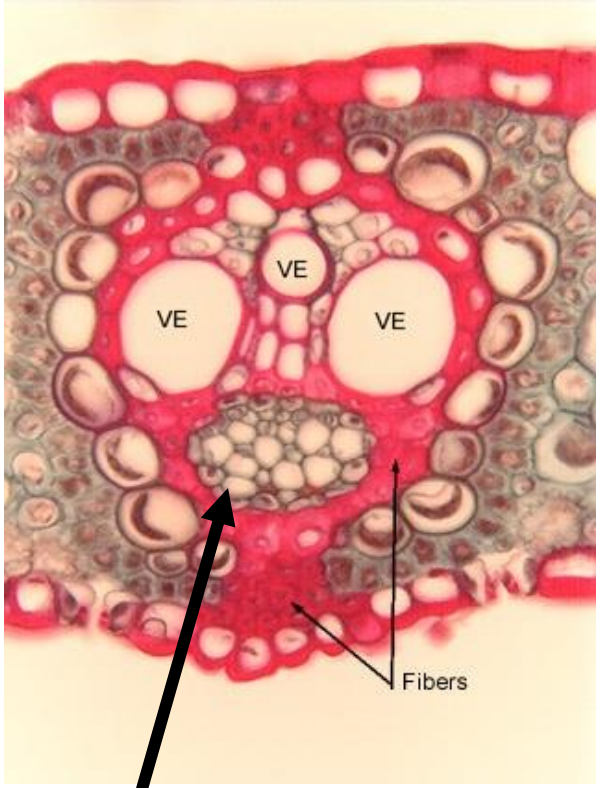


- Xylem is made of tracheids only
- soft wood of pines
- tracheids often have spiral thickenings on them
- Vessels make the wood of flowering trees “hard” wood

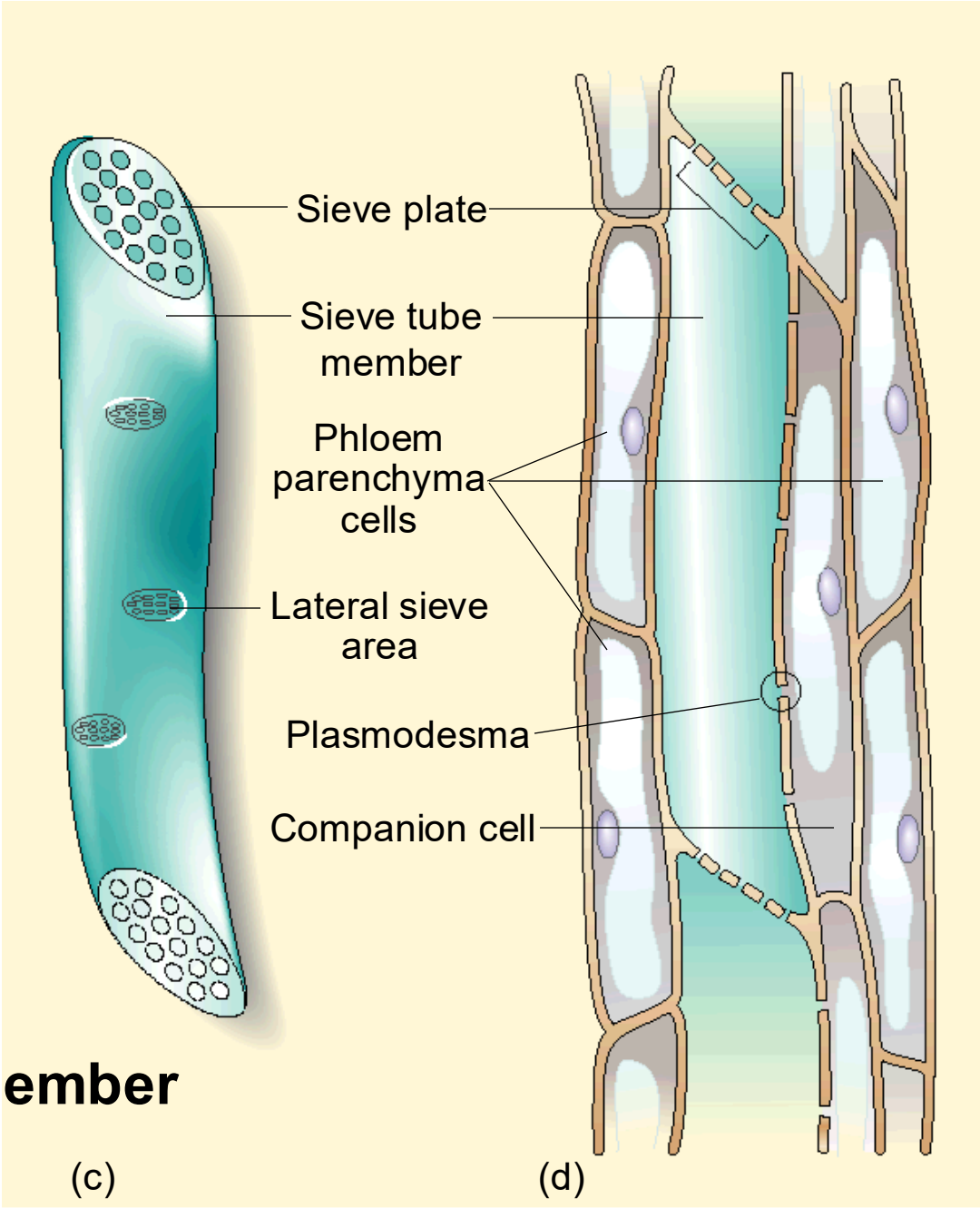
Complex Tissue: Vascular Tissues

- Phloem Tissue
  - transports dissolved food (sugars) through the plant to all its organs

  1. sieve tube members
  2. companion cells
  3. parenchyma
  4. fibers
  5. rays



phloem

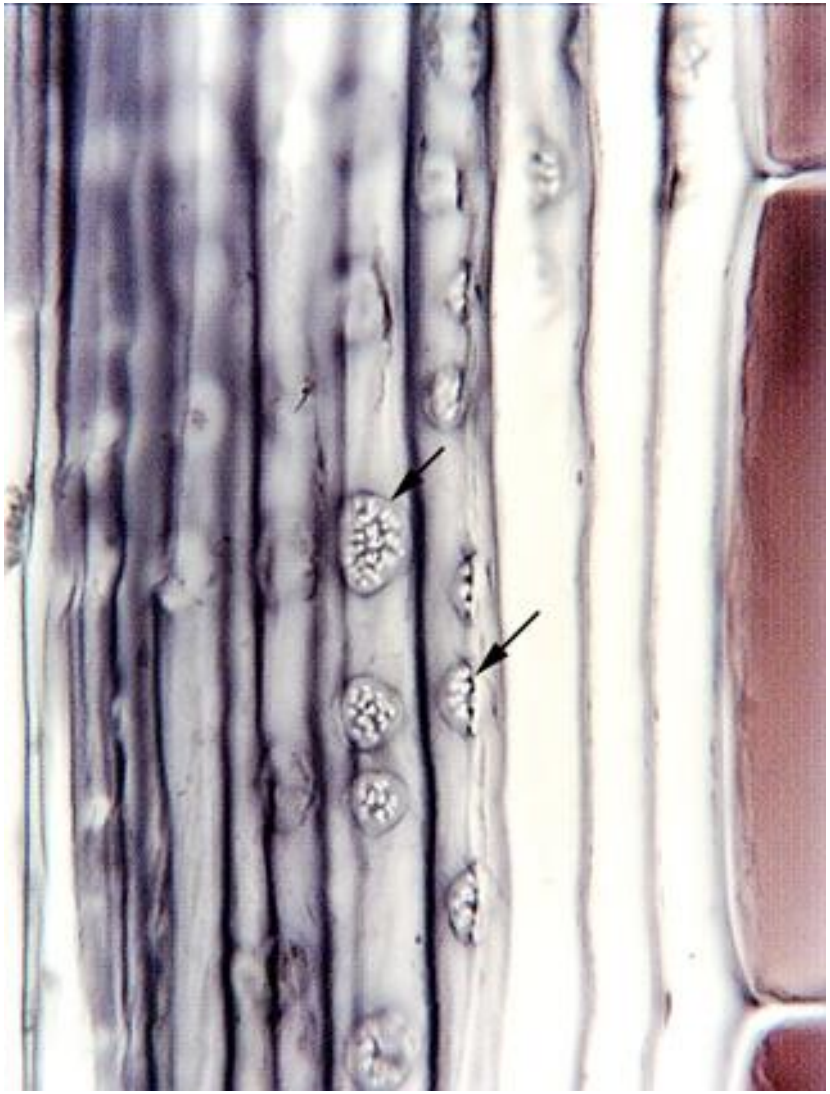


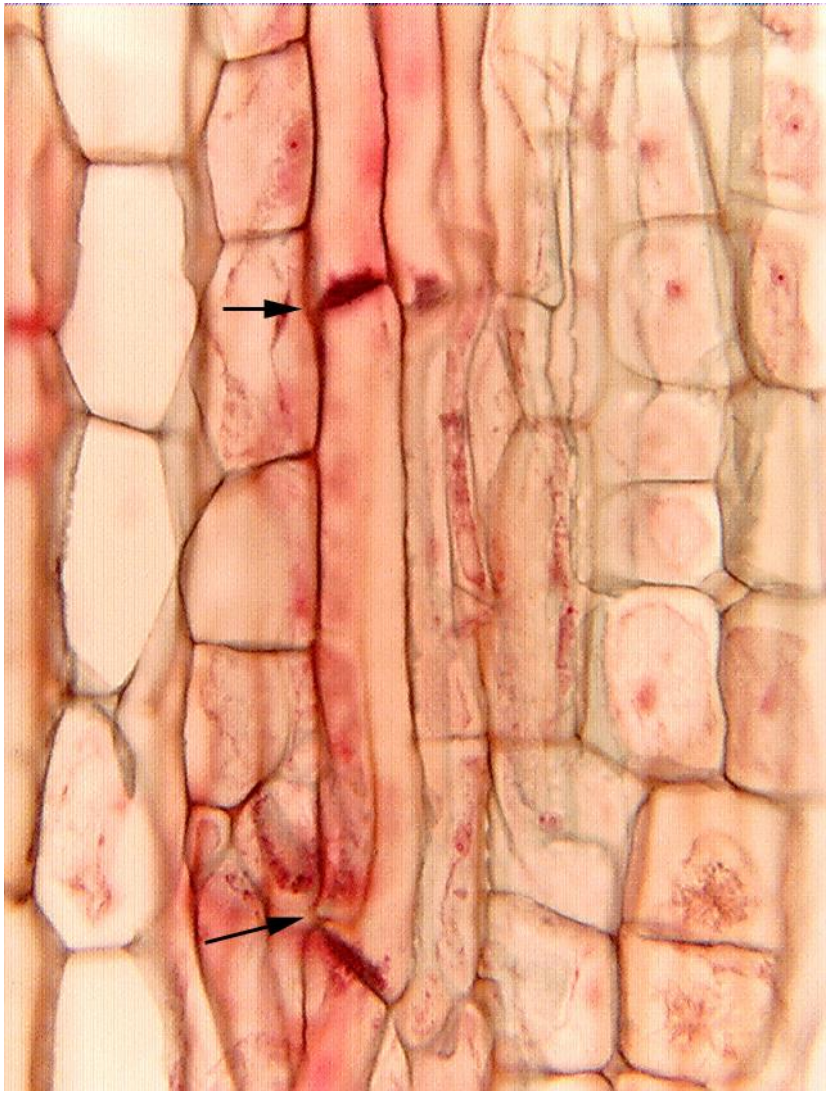
© 2002 Brooks/Cole - Thomson Learning

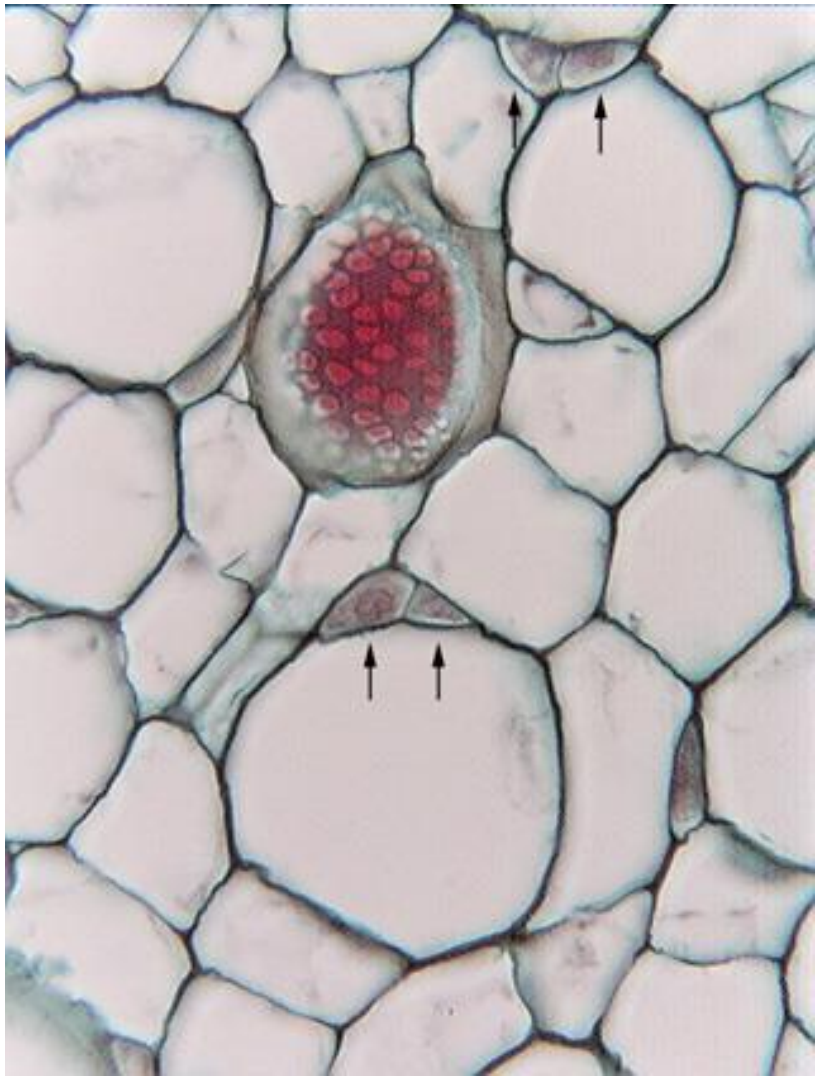
Sieve Tube Member

Phloem tissue

Sieves and Sieve tube cells





Phloem: Companion cells

- Companion cells are “companions” to sieve tube cells
- Because sieve tube cells have no nuclei, the companion cells provide all the nutrients to the sieve tube cell